

NETWORK INTRUSION DETECTION SYSTEM

Suricata IDS 8.0.5

Kali Linux on VMware

59 — Alerts

8 — Rules

5 — Types

59 — Detected

NIDS — Suricata

Network Intrusion Detection System

CodeAlpha Internship Task 4 | Muntaha Ghafoor

59 Real Network Alerts Detected
Using Suricata IDS on Kali Linux VMware
Custom Rules + Python Dashboard

github.com/Muntaha-Ghafoor/NIDS-Suricata

Tools & Environment Setup

Suricata 8.0.5

IDS Engine — monitors eth0 live traffic in real-time. Processes every packet against 8 custom rules and logs alerts to eve.json and fast.log.

Kali Linux

OS platform running on VMware Workstation. Network interface eth0 assigned IP 192.168.137.128/24 used as the IDS sensor.

Nmap 7.99

Used to simulate real-world attacks — SYN port scans, aggressive scans, targeted port scans on 21/22/23/4444.

Python 3 + Matplotlib

Custom dashboard.py script parses eve.json and generates 4 charts: alert types, top IPs, protocols, severity.

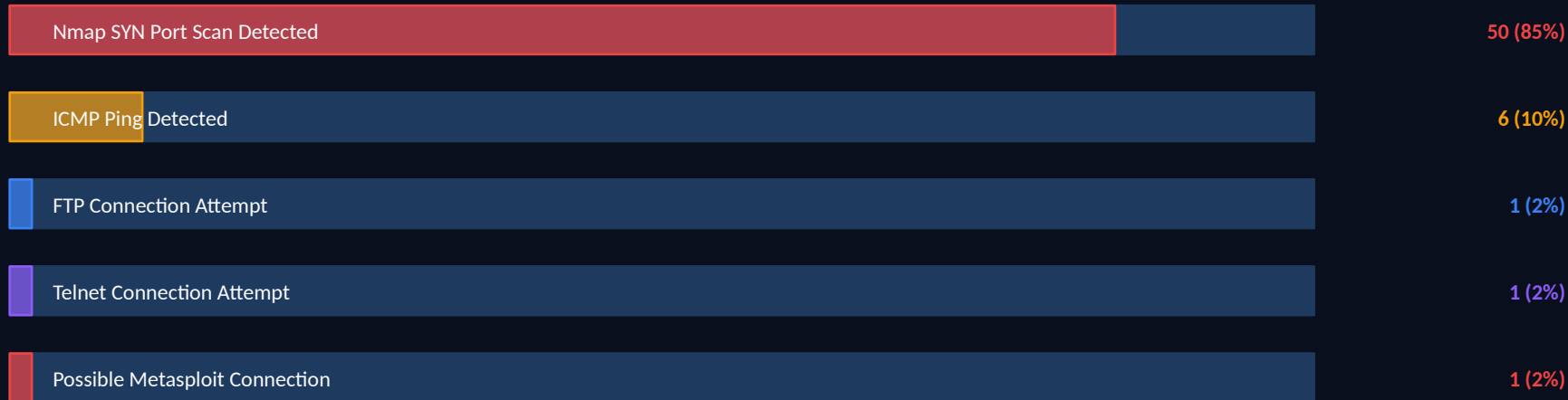
8 Custom Suricata Detection Rules

SID	Rule Name	Proto	Severity
1000001	ICMP Ping Detection	ICMP	MED
1000002	Nmap SYN Port Scan	TCP	HIGH
1000003	SSH Brute Force	TCP	HIGH
1000005	Telnet Connection	TCP	HIGH
1000006	FTP Connection	TCP	HIGH
1000007	DNS Flood	UDP	MED
1000008	Large ICMP Packet	ICMP	MED
1000010	Metasploit Port 4444	TCP	HIGH

Alert Results — 59 Detections



Breakdown



Response Mechanisms & Best Practices

Port Scan

Block attacker IP: `iptables -A INPUT -s ATTACKER_IP -j DROP`

SSH Brute Force

Enable fail2ban, restrict SSH to key-only auth, change default port.

Metasploit (4444)

Isolate host immediately. Kill process on port 4444. Investigate.

FTP/Telnet

Disable legacy protocols. Use SFTP instead of FTP, SSH instead of Telnet.

ICMP Flood

Rate limit: `iptables -A INPUT -p icmp -m limit --limit 1/s -j ACCEPT`

General

Always run Suricata 24/7. Review fast.log daily. Alert SOC team.

Summary & Conclusion

IDS Tool	Suricata 8.0.5
Platform	Kali Linux on VMware
Interface Monitored	eth0 — 192.168.137.128
Custom Rules	8 rules written from scratch
Total Alerts	59 real detections
Attack Types	5 — Scan/ICMP/FTP/Telnet/Metasploit
Charts Generated	4 Python Matplotlib charts
Internship	CodeAlpha — Task 4